

The image displays a sequence of Feynman diagrams representing a mathematical identity. It begins with a sum of four terms, each featuring a vertical red line and two horizontal blue lines. The first term shows a fermion loop on the left with momenta p_1 and p_2 , and a fermion loop on the right with momenta k and $k-p_1$. The second term shows a fermion loop on the left with momenta k and $k-p_1$, and a fermion loop on the right with momenta k and $k-p_2$. The third term shows a fermion loop on the left with momenta p_1 and $k-p_2$, and a fermion loop on the right with momenta p_1 and k . The fourth term shows a fermion loop on the left with momenta $k-p_1$ and $k-p_2$, and a fermion loop on the right with momenta $k-p_1$ and k . The sum of these four terms is equal to the product of two disc diagrams, $\text{disc}_{p_1^2} \text{disc}_{p_2^2}$, which are shown on the right. Each disc diagram is a triangle with a fermion loop on one side and a wavy line on the other.

$$\begin{aligned}
 & \text{Diagram 1} + \text{Diagram 2} + \text{Diagram 3} + \text{Diagram 4} = \text{disc}_{p_1^2} \text{disc}_{p_2^2}
 \end{aligned}$$